

## Linear Circuits 2

### Circuit Element Identification Lab Report

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## Objectives

We are given a concealed circuit with the following specifications:

- The concealed circuit contains two circuit elements only: Element X and Element Y
- The elements are connected in series
- Both elements X and Y can be a resistor, inductor, or capacitor
- The concealed circuit must contain at least one resistor

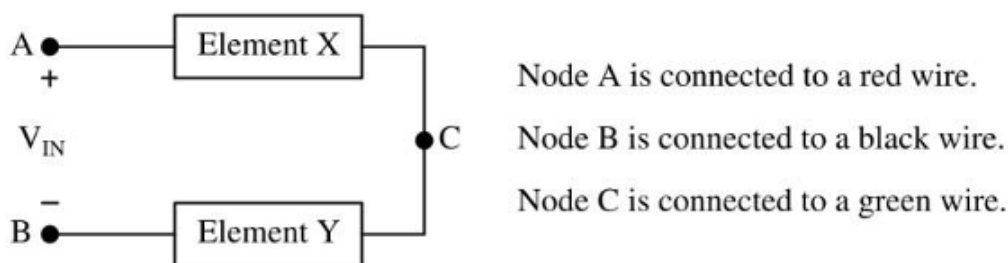


Figure 1: Concealed circuit under test. Node A is connected to a red wire, Node B to a black wire, and Node C to a green wire.

We are given four tasks:

- Identify the type of element that is connected between node A and node C.
- Determine the value for element X. The predicted value must fall within  $\pm 20\%$  of the actual component value.
- Identify the type of element that is connected between node B and node C.
- Determine the value for element Y. The predicted value must fall within  $\pm 20\%$  of the actual component value.

## Dissection of Experimental Plan: Part One

To begin with, the first part of our plan was to determine the combination of elements X and Y. We concluded that there were only three combinations, which include RR, RL, and RC. The first test that we performed was to apply a square wave input to observe the phase difference of the waveforms across both elements X and Y.

## Experimental Results: Part One



Figure 2: Function generator settings: 1 kHz, 10 V<sub>pp</sub> square wave



Figure 3: Oscilloscope output of the waveforms across component X and Y

Using a 1 kHz, 10 V<sub>pp</sub> square wave, we observe the oscilloscope output of component X and Y. Utilizing the oscilloscope, we find that there is no phase difference. As a result, we determine that the concealed circuit has two resistors in series.

## Dissection of Experimental Plan: Part Two

After gaining our Part One results, we determined the concealed circuit was a voltage divider. We had  $V_{ab}$  as the input voltage,  $V_{cb}$  as the voltage drop across resistor Y, and  $V_{ac}$  as the voltage drop across resistor X, equal to  $V_{ab} - V_{cb}$ .

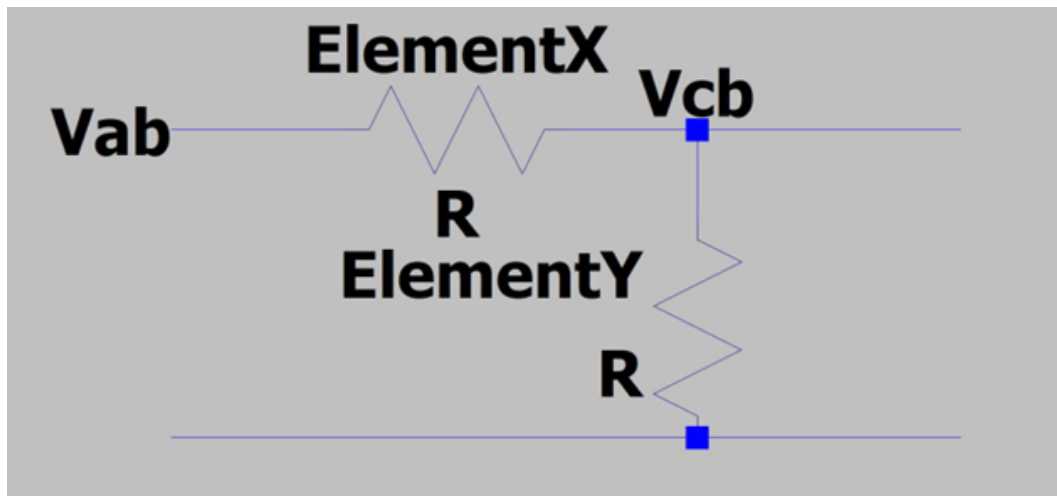


Figure 4: Concealed circuit modeled as a voltage divider, with Element X and Element Y both resistive

We required two equations to determine the values for both resistors: an equation for the voltage drop across resistors in series, and an equation for the equivalent resistance between two series resistors.

$$V_{out} = V_{in} \times \frac{R_y}{R_x + R_y}$$

$$R_x + R_y = \frac{V_{in}}{I}$$

Figure 5: Voltage-divider equation and series-resistance equation

## Experimental Results: Part Two

The second test performed on the concealed circuit was a current and voltage measurement.



Figure 6: Measured AC current,  $I = 0.2204 \text{ mA}$

We measure an AC current  $I$  of  $0.22 \text{ mA}$ .



Figure 7: Oscilloscope voltage and current measurement across the concealed circuit

With  $V_{ab} = 5 \text{ V}$  and  $V_{cb} = 2 \text{ V}$ , we substitute the values into the equation for equivalent resistance between two series resistors:

$$R_x + R_y = \frac{5}{0.22 \times 10^{-3}} \approx 22,727 \Omega$$

$$2 = 5 \times \frac{R_y}{22,727}$$

$$R_y = \frac{2}{5} \times 22,727 \approx 9,091 \Omega \quad R_x = 22,727 - 9,091 \approx 13,636 \Omega$$

Figure 8: Solving for  $R_x$  and  $R_y$  from the measured current and voltages

We concluded that Element X was a 9k $\Omega$  resistor and that Element Y was a 14k $\Omega$  resistor.

AC

$V_{rms} = 1.67V$        $V_{avg} = 5V$

$5 - 1.67 = 3.33V$

1kHz  $T = 1ms$

Phase Angle =  $0.02^\circ \approx 0$

$\cos(0) = 1$  so, circuit is RA  
(purely resistive)

$I_{source} = 0.22mA$        $\frac{5}{0.22 \times 10^{-3}} = 22.7kA$

$\cos(0) = 1$

$22.7k \cos(0) = 22.7k$        $22.7k \sin(0) = 0j$

$Z = 22.7k + j0$

Figure 9: Hand calculations backing the previous results

## Conclusion

We observed that both the input and output signals had no phase difference, by comparing the square wave behavior of  $V_{ab}$  and  $V_{cb}$ . This alerted us to the conclusion that the concealed circuit was an R-R combination. Once we realized it was two resistors in series, we measured the input and output voltages, took note of the current flowing through the concealed circuit, and used the voltage divider and series resistance equations to determine the value of each resistor. Overall,

the results aligned with our expectations, confirming the approach for identifying and solving the concealed circuit worked.